

Break/fix drill and capstone integration

Day 7 - Afternoon

Running scenario: Thara Hospitals patient portal pilot

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CONCEPT 1

The Well-Architected Framework as a review method

The review is the defence.

15 min



IT

HEAD OF IT

I must defend the pilot to the board. The review is the defence.

The Head of IT, on why a design is questioned before it is submitted.

Six pillars as questions, not adjectives



- **Operational excellence**: what happens when nobody is watching?
- **Security**: who can, and how would we know?
- **Reliability**: what fails, and what happens then?
- **Performance efficiency**: is each part the right kind and size?
- **Cost optimization**: can every line be explained?
- **Sustainability**: is anything running that nobody uses?

Back to Day 1



Is any part of the capstone better as a container, and what would that change?

CONCEPT 2



Decommissioning is a skill

Taking a pilot down without losing what must be kept.

15 min

Delete in order, keep on purpose



- Instances first, the network last: the reverse of how they depend
- Keep until marking: the image, the snapshots, the trail and its bucket
- Keep `alias/thara-records`: the snapshot cannot be restored without it
- The proof: a forecast for next month of zero, or near it

If your account hosts the capstone, tick the list and delete after the viva.

The Free Plan clock

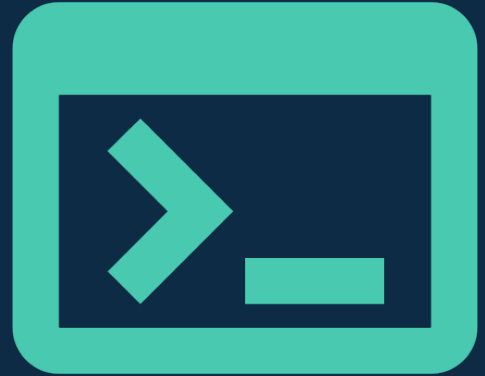


- It ends after **six months**, or when the credits are used
- The account then closes
- AWS keeps its content for **90 days**; an upgrade restores it
- Write your own date in your ledger today

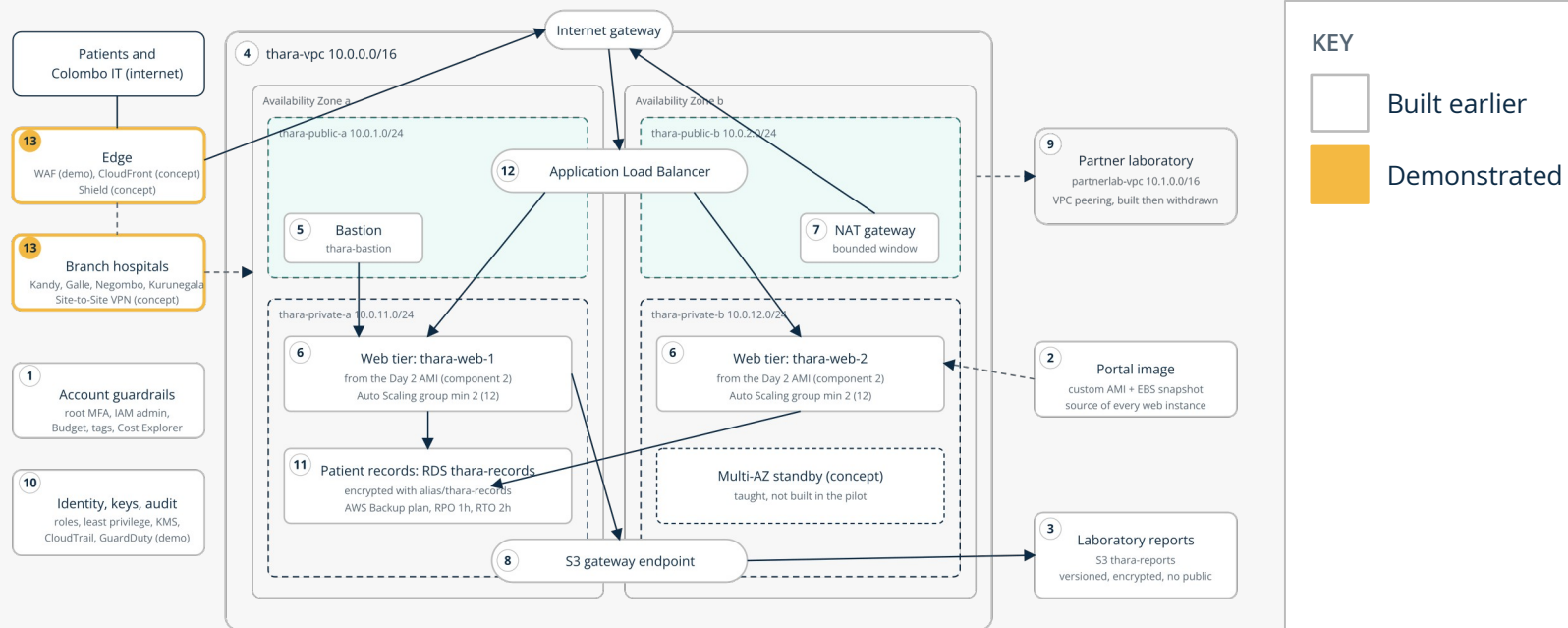
Break/fix drill and capstone integration

Estimated credit use: about USD 0.30

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The Thara architecture



SEGMENT 1 - 20 MIN



Introduce faults in a partner's environment

Cost: nil

Bring it back, and prove it works

13

Start `thara-bastion` and the Day 3 `thara-web-1`. Set the SSH source to My IP. Hop to `thara-web-1` and run both commands.

Expected: The first lists `reports-index.txt`. The second is refused.

```
$ aws s3 ls s3://thara-reports-<student id>
$ aws s3 ls
```

Do not launch another `thara-web-1` from the template.

Swap seats, never passwords

14

Fault 1: remove the default route from `thara-public-rt`. Fault 2: delete every inbound rule of `thara-web-sg`.

Expected: Two ticks on the fault sheet.

14

Fault 3: add a default route to the internet gateway in `thara-private-rt`. Fault 4: turn off Block Public Access on the reports bucket.

Expected: Two more ticks. Add no public statement.

14

Fault 5: an inline policy on `thara-web-role` that allows every action on every resource, under any name.

Expected: Five ticks, and nothing else changed.

Exactly these five. Every one is safe and can be put right.

SEGMENT 2 - 50 MIN



Diagnose and repair your own

Cost: nil

Find all five, and keep the record

15 Repeat the three checks. Then read the route tables, `thara-flow-logs`, CloudTrail event history, Access Analyzer and the console.

Expected: Five rows: symptom, evidence, fix, time.

15 Repair each fault as you find it. Run the three checks again.

Expected: The results of step 13. Two sources told you nothing; write that down.

A silent source is evidence too. Look at the name on every event.

SEGMENT 3 - 20 MIN



Debrief by fault

Cost: nil

Debrief, fault by fault



16

Which evidence source found each fault fastest?

Expected: A fastest source for every row.

16

Which of the five would the Data Protection Officer care about most?

Expected: One sentence from your pair. The room will not agree at first.

Activity: Capstone integration workshop, 60 minutes



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1. Each group presents its diagram to another group for a peer review. The six pillars are the checklist. Ten minutes each way
 2. The Well-Architected Tool is run against one volunteer design
 3. Each group records three changes it will make before submission

Prove it, then break it



PROVE IT

- All five faults repaired
- The symptom of each, written down with its evidence source

BREAK IT

- This afternoon is the break it
- Step 14 holds the faults. Step 15 holds your record

Take your account down



TEARDOWN

- Terminate `thara-web-1` and `thara-bastion`
- Delete `thara-web-lt` and `thara-alerts`. Disable GuardDuty
- Delete `thara-daily` and its recovery points
- Delete the flow log, the endpoint, `thara-db-subnets`, then `thara-vpc`
- Forecast zero, or near it: the final ledger row

Common mistakes



COMMON MISTAKE

Turning off Block Public Access raises a warning.

Nothing warns you. Access Analyzer reports only a real public grant.



COMMON MISTAKE

A wildcard policy is harmless if nobody uses it.

It is the first thing an attacker uses.

Before the review session



READ

The learning outcomes

Revise against the eight outcomes and their sub-outcomes.

PRACTICE

Practice questions 1 to 6

Any that you have not yet attempted.

TASK

Your group

Continue the capstone build for submission after the review session.

Questions & Discussion



Next: the review session.



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